# FUNCTIONAL REQUIREMENTS

**Product Name:** SAUCEDEMO Portfolio Project

**Prepared by:** Larissa Yasmin C. Souza

**Date:** 17th March 2026

# 1. Overview

## 1.1 Project Purpose

The **SauceDemo** application serves as a standardized e-commerce environment primarily used for software quality assurance and automation testing. The core objective of this project is to validate the **end-to-end (E2E) flow**, ensuring the user journey remains seamless and free of critical systemic failures.

## 1.2 Validation Scope

The testing scope covers the entire customer interaction lifecycle. This includes:

- **User Authentication:** Secure system login.
- **Product Management:** Catalog browsing and shopping cart interactions.
- **Checkout Process:** Order completion and data validation.
- **Session Management:** Secure logout and state persistence.

Rigorous validation of these stages is critical to guarantee a fluid and reliable user experience.

# 2. Functional Requirements

The following table details the essential functional requirements for the SauceDemo platform. Each requirement is categorized by description, priority level, and specific acceptance criteria.

| ID | Requirement | Description | Priority | Acceptance Criteria |
|---|---|---|---|---|
| **FR-01** | **User Authentication** | Users must be able to log in using valid credentials. | High | • System redirects to the Inventory page upon success.<br>• Session cookie is securely established. |
| **FR-02** | **Product Inventory** | Users should see a comprehensive list of available products. | High | • Each product must display a name, description, and price.<br>• Product images must load correctly. |

| ID | Requirement | Description | Priority | Acceptance Criteria |
|---|---|---|---|---|
| **FR-03** | **Shopping Cart** | Users should be able to add or remove items from the shopping cart. | High | • Cart badge updates in real-time.<br>• Item persists in the cart during navigation. |
| **FR-04** | **Checkout Process** | Users must complete a shipping information form to finish the purchase. | High | • Error message appears if First Name, Last Name, or Postal Code is missing.<br>• Order summary displays correct total before finishing. |
| **FR-05** | **User Logout** | Users must be able to securely terminate their session. | Medium | • User is redirected back to the Login page.<br>• Access to internal pages is restricted after logout. |

## 2.1 Additional Considerations

All requirements classified as **High Priority** constitute the application's **critical path (Happy Path)** and must be prioritized during both automated and manual test cycles. Although the **Logout** requirement is classified as **Medium Priority**, it is fundamental to ensure user session security and must be validated at the conclusion of every purchase flow.